

Project:

Analysis & Dark Matter Physics Simulation for the Dark Photon

Team Members:

- Nikhil Chaba, nchaba2021@my.fit.edu
- Samuel Rock, srock2023@my.fit.edu
- Jacob Woods, jwoods2022@my.fit.edu

Faculty Advisor:

- Pietro Iapozzuto, piapozzuto2015@my.fit.edu

Clients:

- Dr. Marcus Hohlman - Physics Professor
- Pietro Iapozzuto - Graduate Student with Dr. Hohlman
- Other Graduate Students

Progress of Current Milestone (Matrix):

Task	Completion %	Nikhil	Sam	Jacob	To do
Understanding the Weizsacker-Williams approximation codebase	95%	-	1/2	1/2	We still need to confirm this is the correct and wanted approach from our client/professor.
Do a gold e- collision analysis in MadGraph and compare it to $p e^- \rightarrow p e^-$ collision	40%	1/3	1/3	1/3	We have the initial simulation running of a gold collision, but we want to be able to run our scripts on the data output to format the data in a readable way.
Write a User Manual	10%	1/3	1/3	1/3	We have to write the processes and steps required for later parts of our updated pipeline

Testing the Weizsacker-Williams approximation codebase	0%	1/3	1/3	1/3	We will still need to confirm this is the correct approach before implementing this ourselves.
--	----	-----	-----	-----	--

Discussion of each accomplished task (and obstacles) for the current Milestone:

- **Task 1:** Understanding the Weizsacker Williams approximation code involved documenting the codebase line by line in a Google document. The challenge came when trying to understand what the physics equations represented. As computer scientists, we were able to identify what the ideas and structure behind the code were, but when it came to what the algorithm was calculating, we handed that off to the physics professionals we are working with.
- **Task 2:** We have managed to have MadGraph run and create a gold/electron collision simulation. However, the raw output is unobservable for our research, and we need our secondary software to run on it and produce a readable output for us. That is where the challenge comes in.
- **Task 3:** Just started writing the user manual
- **Task 4:** Testing the codebase comes into play once we know for sure that this will be our correct approach in solving our pipeline mathematical problem.

Discussion of the contribution of each team member to the current Milestone:

- Nikhil Chaba:
- Samuel Rock: Wrote part of the Milestone 4 documentation and presentation. Documented about half portion of the Weizsacker-Williams approximation codebase. Researched running Pythia8 scripts on outputs from a gold/electron collision to understand the raw output values.
- Jacob Woods: Wrote part of the Milestone 4 documentation and presentation. Documented about half portion of the Weizsacker-Williams approximation codebase. Researched running Pythia scripts on outputs from a gold/electron collision to transform into readable output.

Plan for the next Milestone (Matrix)

Task	Nikhil	Sam	Jacob
Confirm that the Weizsacker algorithm is the correct path for us to follow.	-	1/2	1/2

Testing the Weizsacker-Williams approximation codebase (contingent on the confirmation that this is the correct path)	1/3	1/3	1/3
Analyze a gold e- collision at the reconstruction/root level and compare it to p e- > p e- xp collision	1/3	1/3	1/3
Write the user manual	1/3	1/3	1/3

Discussion of each planned task for the next Milestone:

- **Task 1:** Confirming whether we pursue the implementation of the Weiszacker-Williams approximation codebase will involve a discussion with our client (Professor Hohlmann). This is important because it is possible that we follow another route.
- **Task 2:** Given that we have confirmed this is correct, we will then download the code and run it within our environments.
- **Task 3:** We will run the collision simulation and then run the raw output through our own software.
- **Task 4:** Continue writing the user manual, focusing on the latter parts of our updated pipeline

Date(s) of meeting(s) with Client during the current milestone:

9/28

Client feedback on the current milestone

- x

Date(s) of meeting(s) with Faculty Advisor during the current milestone:

9/3, 9/10, 9/28

Faculty Advisor feedback on each task for the current Milestone

- Task 1:
- Task 2:
- Task 3:
- Task 4:

Faculty Advisor Signature: _____ Date: _____